Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/15/2005
PI Name:	Asher, Sanford Ph.D.		
Project Title:	Development of Intelligent Polymerized Crystalline Colloidal Array Sensor for Invivo Detection of Cancer Marker Proteins		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:	National Institutes of Health/National Cancer Institute	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pittsburgh	**State:**	PA
Zip Code:	15260	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/01/2002	**End Date:**	11/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	-----		
COI Name (Institution):			
Grant/Contract No.:	NAS2-02058		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our research and development program will create a biomolecular sensor technology platform to enable non-invasive or minimally invasive early detection of signatures of disease, especially cancer. We will utilize our recently invented intelligent polymerized crystalline colloidal array (IPCCA) material for in vivo sensing. These IPCCA materials and devices utilize novel molecular recognition chemistries within materials containing periodic nanostructures of composite nanoparticles. The signal generation utilizes diffraction of incident light by the nanoparticle arrays. This diffraction of light serves as the interface between in vivo detection and targeted intervention. This technology platform can be multifunctional, in that different PCCA sensors for different disease signatures can be fabricated by utilizing different molecular recognition elements in these biocompatible PCCA materials To develop implantable sensors for prostate specific antigen and bladder cancer antigen proteins (BLCA4). xCancer is one of the most prevalent dieseases troubling		

	society today. Cancer can be detected earlier, especially in cetain risk groups, or remission monitored more easily if a minimally invasive technique can be designed to monitor the body. The PCCA will be an implantable detector wihch will visually or with the use of a small spectrophotometer report on the concentration. Successful development of sensors for marker proteins for prostate and bladder cancer will prompt an extensive search for marker proteins specific to toher cancer types. Hopefully, success in this project will allow for earlier detection and easier, less-invasive methods for monitoring cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Novel Sensor motifs to better detect the early presence of cancer would have a major impact on cancer diagnosis and treatment.
Task Progress:	Important observations have been achieved in understanding how to implement implantable cancer sensors.
Bibliography Type:	Description: (Last Updated: 02/27/2004)